
O-RAN Work Group 10 (OAM for O-RAN)

Application Packaging General Aspects & Principles

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Foreword

This Technical Specification (TS) has been produced by WG10 of the O-RAN Alliance.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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- xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.
- yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.
- zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the general aspects, principles, use cases, information model, and requirements pertaining to the format and content of an Application Package which then allows the Network Operator to Onboard the Solution Provider's Application to the SMO through the Onboarding SMOS.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

- [1] O-RAN.WG11-O-RAN-Security-Requirements-Specification: "O-RAN Security Requirements"
- [2] O-RAN.WG10.TS.Information Model and Data Models, "O-RAN Information Model and Data Models Specification"
- [3] O-RAN.WG10.TS.O1-Interface, "O1 Interface Specification"
- [4] O-RAN WG1.OAD: "O-RAN Architecture Description".

2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

- [i.1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [i.2] O-RAN WG10.OAM-Architecture: "O-RAN OAM Architecture"
- [i.3] O-RAN-WG6.ORCH-Use-Cases: "Cloudification and Orchestration Use Cases"
- [i.4] O-RAN-WG11.AppLCM-Security-TR: "Security for Application Lifecycle Management"

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms [given in [i.1] and the following] apply:

Application: The software aspect of a vendor product for a Network Element [i.2], an xApp or rApp which is sold/delivered to an operator.

Accounting Descriptor: An artifact which describes any license and/or usage records used to identify measured license accountability.

Alarm Dictionary: An artifact which describes all alarms which can be emitted by the Application.

Application Artifact: A file contained within an archive such as the Application Package.

Application Management Package: The set of information provided by a Solution Provider to a Service Provider in a dedicated package containing only Management Descriptors.

Application Package: The set of artifacts provided by a Solution Provider to a Service Provider representing the Application as a whole.

Network Element: The definition of a Network Element (NE) is given in 3GPP TS 21.905 [i.1] clause 3.N.

NF Deployment: Refer to the Cloudification and Orchestration Use Cases [i.3].

Network Function: A standardized set of functionality which is used to implement the RAN.

Management Descriptor: A descriptor describing some aspects for the FCAPS management for a Network Function.

Onboarding: The process of validating and ingesting an Application Package to make it available to other services within the SMO.

Performance Dictionary: An artifact which describes all performance measures which can be collected by the Application and sent to a management system according to its subscription criteria.

Security Descriptor: An artifact which can describe needed information for the Security Aspects of FCAPS for the Application.

Service Provider: The RAN service provider often referred to as the operator.

Software Package: A binary image to be downloaded to a PNF as some or all of its software build.

Solution Provider: Application development entity often referred to as a vendor.

Telemetry Schemas: Artifacts which describe the schema for telemetry data sent from the Application to a management system.

3.2 Symbols

Void

3.3 Abbreviations

For the purposes of the present document, the abbreviations [given in [i.1] and the following] apply:

AAS	Active Antenna System
CNF	Cloudified Network Function

eNB	e NodeB (applies to LTE)
gNB	g NodeB (applies to NR)
JSON	JavaScript Object Notation
NE	Network Element
NF	Network Function
NRM	Network Resource Model
O-DU	O-RAN Distributed Unit
O-RU	O-RAN Radio Unit
PNF	Physical Network Function
SDO	Standards Development Organization
SMO	Service Management & Orchestration
YAML	Yet Another Markup Language
XML	eXtensible Markup Language

4 General aspects of application packaging

4.1 Aspect: artifact management

4.1.1 Original artifacts

Artifacts are expected to be preserved in the original, as-is format, received from the vendor. This is not only crucial for artifact integrity checks but also for audit of what was received. Artifacts may be duplicated and modified by later lifecycle operations. Those modified artifacts may need to be compared to the original artifact when troubleshooting is needed.

4.1.2 Artifact security

All artifacts contained or referenced within the package shall be identified in a manifest artifact. All artifacts in the package and external references identified in the manifest are required to be checked for authenticity and integrity according to O-RAN Security Requirements Specification [1].

4.1.3 Artifact references to other artifacts

4.1.3.1 Local relative reference

Artifacts can reference other artifacts within their content. This can be done using a relative address local to the root artifact. It is expected that the Onboarding SMOS can resolve a local relative reference to the absolute path of an onboarded artifact.

4.1.3.2 External URL reference

Artifacts can reference another using a URL which is external to the Onboarding Package. It is expected that the Onboarding SMOS can resolve a external URL reference to the absolute path of an onboarded artifact.

4.1.4 Package directories

4.1.4.1 Standardized directories

Some directories in the Onboarding Package may be standardized, and certain types of artifacts may be required to be placed in those

4.1.4.2 Vendor extended directories

It is expected that a vendor may further organize their data within the standardized directories. Therefore, a vendor is free to add any sub-directories they deem fit to the standardized directories.

5 Principles of application packaging

5.1 Principle: data minimization

5.1.1 Data reduction

Generally, it is not desirable to replicate data. Anytime data is replicated a source-of-truth needs to be identified and consistency of the copy comes into question. Additionally, depending on the size of the copied data the size of the overall object can be drastically increased as there will be one or more copy of that data in the entire data set. The copy may also be transformed into a different format. Such a translation may inject discrepancies or inaccuracies into the copy while also requiring consumers to have different translators for the copies. Therefore, the ideal situation is that there never is any duplication of data.

5.1.2 Data caching

There are cases where data replication adds value. If the format of the source of truth is opaque to the different consumers, then it may be advantageous to put the burden of translation on the supplier. Another scenario is one of latency, if the access to the source-of-truth takes too long for the use case then the data may be replicated as a cache. Techniques to ensure accuracy of the cache when the source-of-truth is modified are usually required.

5.1.3 Data replication balance point

Data Reduction which demands duplication elimination, and data caching which demands data replication are at odds with each other. Therefore, the balance point between them is based on the use cases which use any potential replica of the data.

6 Use cases of application packaging

6.1 Use case conventions

6.1.1 Overview

The present document follows the Use Cases methodology described in clauses 6.1.2, 6.1.3, 6.1.4 and 6.1.5.

Typical use cases are comprised of actors who interact with each other passing data. Those interactions are often translated into functional requirements. In the case of application packaging the requirements are non-functional and there are no interactions. The interactions occur in the lifecycle which have data demands. Those interactions are described in the O-RAN OAM Architecture [4]. This does not mean that there are not scenarios in which analysis can be performed to identify the data requirements. Therefore, this specification has adopted the following conventions to document a scenario, or use case, in a relevant manner to drive the creation of the non-functional requirements for application packaging. The Use Cases can be merged into a composite but for the sake of requirements analysis all their combinatorial variations are not needed and therefore may not be represented.

6.1.2 Use case description

This is the first subclause (6.2.1) in the use case. It is written in natural language. It sets the context for an onboarding activity. It provides enough detail to describe the scenario, set its scope and identify how it is different to other scenarios.

6.1.3 Use case analysis

The "Analysis" is the second (6.2.2) subclause in the use case. It describes why certain information elements are required and why some are not required. Some information elements can be optional, however there is no attempt to address requirements covered by a different scenario.

6.1.4 Use case information model relationships

The Information Model Relationships is a UML Class diagram and is the third subclause (6.2.3) of a use case. It describes the information elements of the application package but does not describe any artifact in the package.

6.1.5 Use case requirement traceability

Requirements are the goal and fourth (6.2.4) subclause of a use case. The section identifies applicable non-functional requirements which are defined in clause 7.2 of this specification. The traceability is a table where each requirement can be associated with an optional comment to clarify its specific applicability to the use case

6.2 Cloudified NF with NF Deployment decomposition: packaging use case

6.2.1 Description

6.2.1.1 Initial package delivery

A solution provider has implemented an O-RAN NF, e.g. an O-CU-CP, consisting of three sets of microservices. Each set contains one or multiple microservice and each set has an independent life cycle. Therefore, the solution provider has decomposed the NF to be realized with three NF Deployments.

In addition, for the management of the NF via the O1 interface the solution provider has prepared a set of management descriptors including NF configuration modules, an alarm dictionary and a performance dictionary.

In relation to the management descriptors, two sub-cases are considered:

- Disaggregated management information: information in the management descriptors is broken down into parts, where each of the part is very tightly coupled with a particular NF Deployment. In other words, if the software of that particular NF Deployment changes, it is likely that the related part of the management descriptor also has to change while other parts of the management information will likely not need to.
- Centralized management information: information in the management descriptors is kept undivided in one single artifact for each of the management descriptor types – configuration, alarm dictionary, performance dictionary- e.g. because it is not tightly coupled to a particular NF Deployment.

6.2.1.2 New NF Deployment software or artifacts with new functionality

The solution provider updates one of the sets of microservices that make up the NF in order to enhance the functionality of the NF. Therefore, it creates a new version of the corresponding NF Deployment containing new versions of software images, Helm charts and of the NF Deployment descriptor.

In addition, in this particular case the management information is affected by the functionality enhancement introduced by the software change. This may not always be the case, i.e. changes in the software of the NF Deployments don't necessarily imply a change in the management information. If the software change affects the management information, depending on the mode used to deliver the management information, it results in a different impact:

- With the disaggregated management information only the part of the management information related to the updated NF Deployment is affected.
- With the centralized management information, as it is not decomposed in parts, a new version of the entire management descriptor is required.

6.2.1.3 New NF Deployment software with a security patch

The solution provider updates a software image in one of the NF Deployments in order to address a security issue. Therefore, it creates a new version of the corresponding NF Deployment containing the new versions of the software image, Helm chart and NF Deployment descriptor.

In this case the management information is not affected by the security patch so there is no need to deliver again the management information.

6.2.2 Analysis

From the use case described in the previous clause the following types of packages are identified:

- A Cloudified NF package
- An NF Deployment package
- A Management package

Where the Cloudified NF package contents are:

- A manifest listing package contents and referencing NF Deployment packages and a Management package. The latter is needed only in the case of centralized management information
- A Cloudified NF descriptor
- Signatures

The contents of the NF Deployment package are:

- A manifest listing package contents and URLs to external artifacts, if any
- An NF Deployment descriptor (e.g. an ASD)
- Software images
- Cloud native descriptor files, e.g. Helm charts
- Management descriptors (configuration modules, alarm dictionary, performance dictionary). These are only needed in the case of disaggregated management information.
- Signatures

The contents of the Management package (only needed in the case of centralized management information) are:

- A manifest listing package contents
- Management descriptors (configuration modules, alarm dictionary, performance dictionary)
- Signatures

6.2.3 Information model relationships

The information model relationships for the Cloudified NF package, the NF Deployment package and the Management package is shown in 6.3.3.

6.2.4 Requirement traceability

Table 6.2.4-1 lists the requirements in clause 7.2.6 that are derived from this use case.

Table 6.2.4-1 Requirements derived from this use case

Requirement	Comment
REQ-CNF-CON-01	
REQ-NFD-CON-01	
REQ-CNF-CON-02	Requirement applicable for the manifest both in the Cloudified NF package and in the NF Deployment package.
REQ-CNF-CON-03	
REQ-NFD-CON-02	
REQ-NFD-CON-03	
REQ-NFD-CON-04	
REQ-CNF-CON-04	
REQ-CNF-CON-05	
REQ-CNF-CON-06	

6.3 Cloudified NF without NF Deployment decomposition: packaging use case

6.3.1 Description

6.3.1.1 Initial package delivery

A solution provider has implemented a logical NF which consists of an aggregation of O-RAN NFs, for example a gNodeB which includes the functionality of a O-CU-CP, an O-CU-UP and an O-DU. In this case the whole NF is realized with a single NF Deployment.

In addition, for the management of the NF via the O1 interface the solution provider has prepared a set of management descriptors including NF configuration modules, an alarm dictionary and a performance dictionary.

6.3.1.2 New Cloudified NF software

The solution provider needs to update the software. Since it is not divided into multiple NF Deployments, the complete software, including software images and Helm charts, and deployment descriptor, has to be delivered for the upgrade.

In this particular case, the management information is not affected by the software change.

6.3.2 Analysis

From the use case describe in the previous clause the following packages are identified:

- A Cloudified NF package
- A Management package

Where the Cloudified NF package contents are:

- A manifest listing package contents and URLs to external artifacts, if any, and referencing a Management package.
- A Cloudified NF descriptor
- An NF Deployment descriptor (e.g. an ASD)
- Software images
- Cloud native descriptor files, e.g. Helm charts
- Signatures

The contents of the Management package are:

- A manifest listing package contents
- Management descriptors
- Signatures

6.3.3 Information model relationships

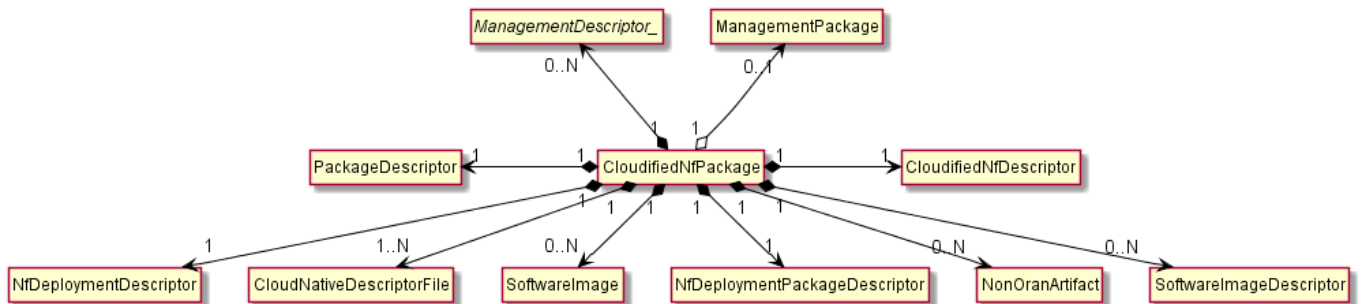


Figure 6.3.3-1 Cloudified NF Package Relationships: monolithic Cloudified NF Package

Figure 6.3.3-1 shows the composition and relationships of a Cloudified NF Package for the case where all software is delivered as part of the Cloudified NF Package, without any dependent NF Deployment Packages. This case only supports a Cloudified NF with one single NF Deployment.

The Cloudified NF Package contains a Package Descriptor, i.e. a manifest, and a Cloudified NF Descriptor. It may contain non O-RAN specified artifacts, e.g. a ReadMe file.

The Cloudified NF Package in this case contains also the artifacts of one NF Deployment (one NF Deployment Package Descriptor, one NF Deployment Descriptor, one or more Cloud Native Descriptor Files, and zero, one or multiple Software Images). The NF Deployment Package Descriptor (the manifest of the NF Deployment) is required as the consumers of the Onboarding SMOS need to know the list of artifacts that make up the NF Deployment.

If the Cloudified NF Package contains Software Images it also contains one Software Image Descriptor for each Software Image. Whether the Software Image Descriptor is realized as a separate artifact or its contents are part of the Package Descriptor (the manifest) is not determined in the information model.

The Cloudified NF Package can contain Management Descriptors or refer to a dependent Management Package.



Some or all the NF Deployment Packages contain in this case Management Descriptors.

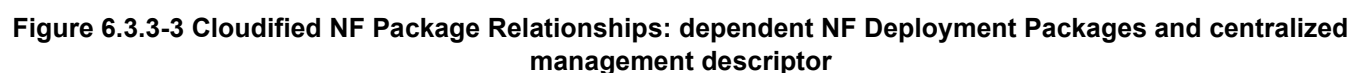


Figure 6.3.3-3 shows the composition and relationships of a Cloudified NF Package and NF Deployment Packages for the case where software delivery is distributed across one or more NF Deployment Packages and the management descriptors are centralized, either in the Cloudified NF Package or in a separate Management Package. This case supports a Cloudified NF with one or multiple NF Deployments.

The Cloudified NF Package contains a Package Descriptor, i.e. a manifest, and a Cloudified NF Descriptor. It may contain non O-RAN specified artifacts, e.g. a ReadMe file.

The Cloudified NF Package in this case refers to one or multiple NF Deployment packages. An NF Deployment Package contains an NF Deployment Package Descriptor (a manifest), an NF Deployment Descriptor, one or more Cloud Native Descriptor Files, and zero, one or multiple Software Images. It may also contain non O-RAN specified artifacts.

If the NF Deployment Package contains Software Images it also contains one Software Image Descriptor for each Software Image. Whether the Software Image Descriptor is realized as a separate artifact or its contents are part of the NF Deployment Package Descriptor (the manifest) is not determined in the information model.

The Cloudified NF Package can contain Management Descriptors or refer to a dependent Management Package.

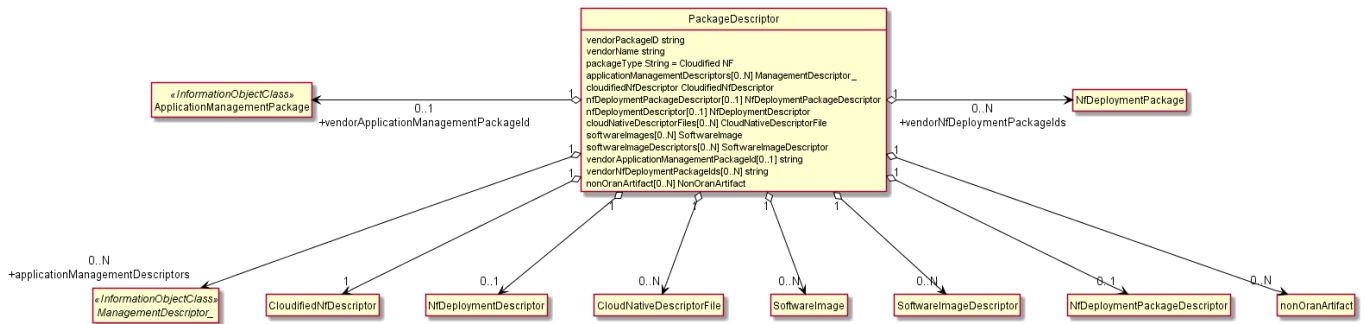


Figure 6.3.3-4 Information Model for a Package Descriptor of a Cloudified NF Package

NOTE: This information model does not prevent a data model in which the SoftwareImageDescriptor is contained in the Package Descriptor.

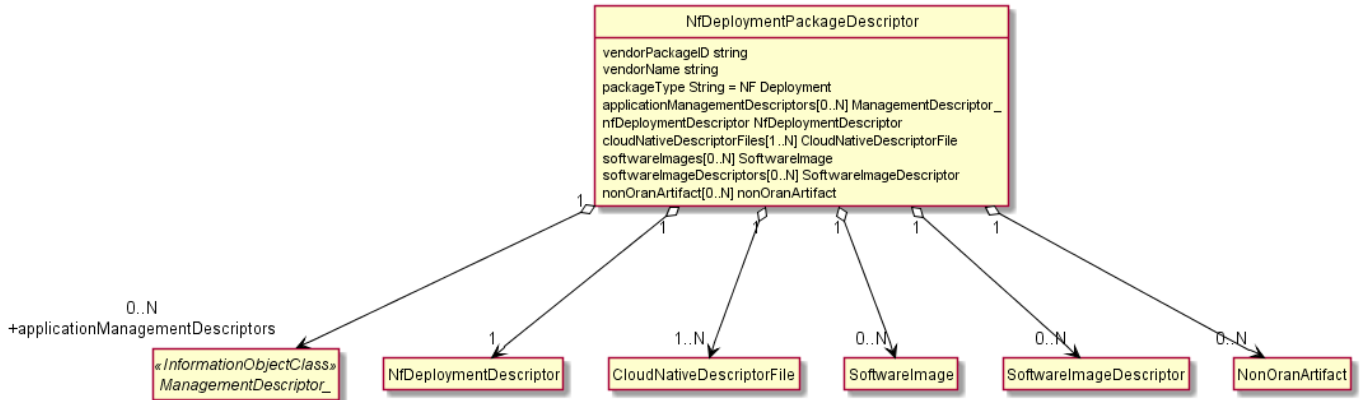


Figure 6.3.3-5 Information Model for a Package Descriptor of an NF Deployment Package

NOTE: This information model does not prevent a data model in which the SoftwareImageDescriptor is contained in the Package Descriptor.

6.3.4 Requirement traceability

Table 6.3.4-1 lists the requirements in clause 7.2.6 that are derived from this use case.

Table 6.3.4-1 Requirements derived from this use case

Requirement	Comment
REQ-CNF-CON-01	
REQ-CNF-CON-02	
REQ-CNF-CON-04	
REQ-CNF-CON-05	
REQ-CNF-CON-06	

6.4 PNF application: packaging use case

6.4.1 Description

The PNF is a physical network appliance containing one or more Network Functions. Life cycle of the device is managed by the SMO as well as provisioning and assurance of the device. A PNF Application Package must be onboarded to create an inventory record which is a pre-requisite for PNF Registration. Detection occurs when a registration message is received from the PNF after it is powered on. To create that inventory record, a PNF Application Package with its management descriptors must be onboarded. The management descriptors are included in the PNF Application package or in one or several of the dependent packages, referenced from the PNF Application package.

The PNF can be delivered either as a single package, or it can be composed of several related packages enabling independent life cycles for PNF components. This use case provides the analysis for data requirements for the PNF Application Package. The contents of an Application Management Package are documented in a separate use case.

6.4.2 Analysis

As depicted in figure 6.4.3-1 the PNF Application Package is an archive containing a manifest identifying all contained artifacts and dependent packages. Delivery of the PNF application software can be done in a single package, or in a component manner using references to PNF Component Packages. This enables independent life cycle management of the PNF components.

The Software Package is an artifact which is the binary data to be provided to the PNF as its software.

Both the PNF Application Package and the PNF Component Package may contain Software Packages, but the PNF Component Package only contains one.

Artifacts are files in the PNF Application Package or in the PNF Component Package, e.g. the package manifest, a SoftwarePackage, etc.

The PNF does require Management Descriptors, such as data to inform the SMO what security settings need to be applied to the hardware after its registration. These can be included in the PNF Application Package, distributed across the PNF Component Packages, or in an independent Application Management Package, see figure 6.4.3-2. As the PNF must have at least one Software Package, there shall also be at least one Software Package Descriptor.

6.4.3 Information model relationships

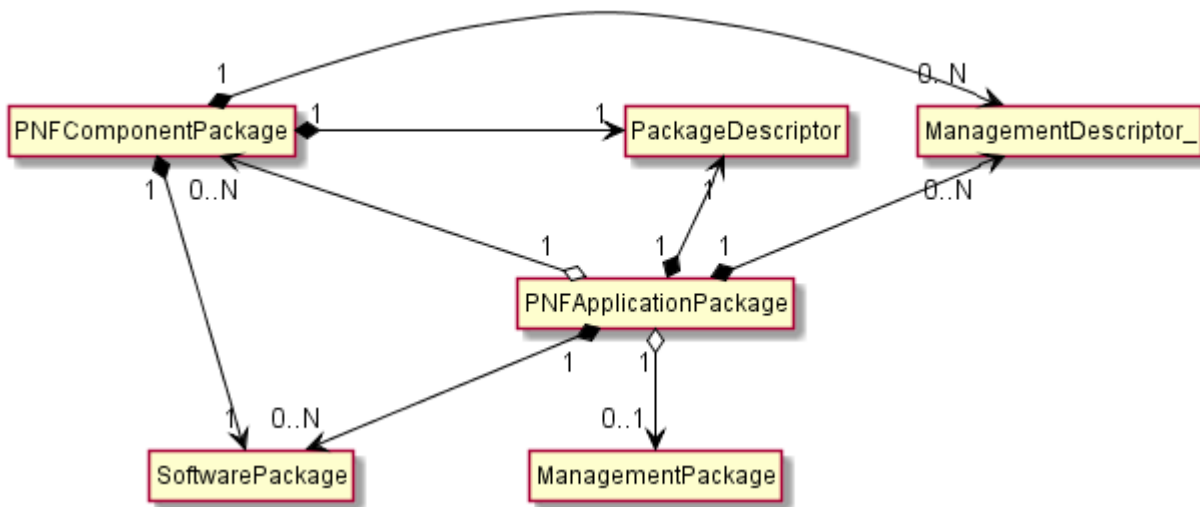


Figure 6.4.3-1 PNF Application Package Relationships

From the relationships depicted in Figure 6.4.3-1 we see that A PNF Component Package contains part of the software of the PNF. A PNF Component Package is limited to only one Software Package. It cannot refer to other packages. The PNF Application Package, in contrast, can contain more than one Software Packages. The PNF Application Package can refer to many PNF Component Packages and it can reference a dependent Application Management Package if desirable.

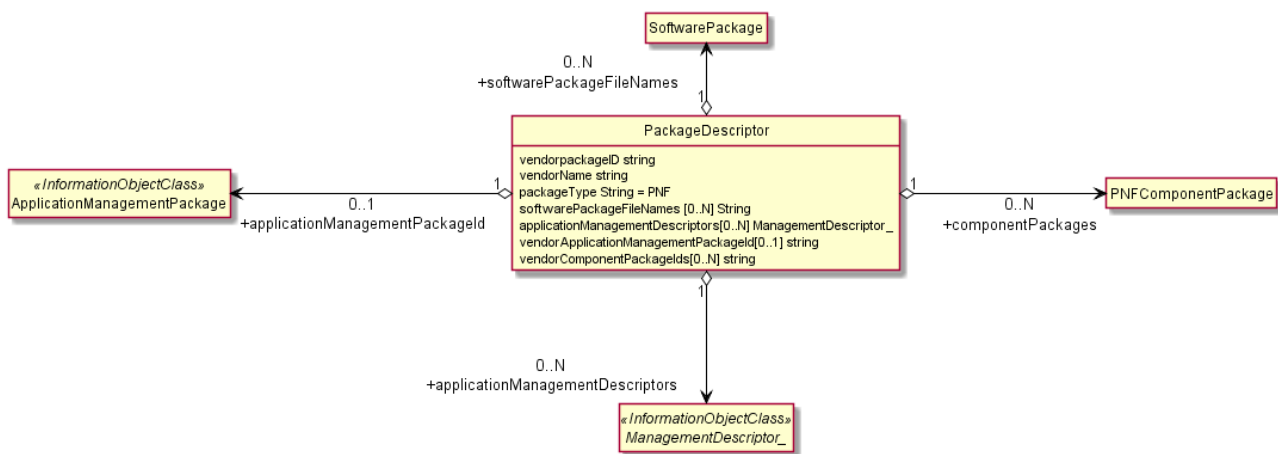


Figure 6.4.3-2 Information Model for a PNF Application PackageDescriptor

6.4.4 Requirement traceability

Requirement	Comment
REQ-COM-CON-01	
REQ-PNF-CON-01	
REQ-PNF-CON-02	

6.5 NF management application: packaging use case

6.5.1 Description

The NF Management Application Package use case is about what information is required to manage a Managed Element based on what it can expose over the O1 interface. The use case focuses on when a management application package referenced by the main Application Package is delivered. However, the Management Descriptors may also be provided in the Application Package or across several dependent application component packages.

6.5.2 Analysis

The Management artifacts are used by the SMO to enable operational management of a NF, rApp, or xApp. In this version of this document only the management of NFs is supported. It is directly focused on those artifacts used to manage the application over the O1 or FH M-Plane interfaces. If a license is required to operate any contained management functions or usage data is required for validation that the volume of any contained NF does not exceed its licensed limits, then an accounting descriptor may be supplied. The Security Descriptor is optional in that elements may not exist and RBAC values and assignments can be discovered once the element has been deployed.

6.5.3 Information model relationships

6.5.3.1 Management descriptors

Elements of the NF Management package can be included in other packages. There is no one Management Descriptor but multiple types of Management Descriptors. An abstract class is used to identify the common data that is required in each Management Descriptor. Figure 6.5.3.1-2 shows the inheritance relationships between the different types of Management Descriptors and the abstract class. Each type of specific Management Descriptor is independently described in the following clauses. The diagrams in this section use a convention of yellow as mandatory and light blue as optional. See Figure 6.5.3.1-1 as the diagramming legend.



Figure 6.5.3.1-1 Relationship UML Color Scheme

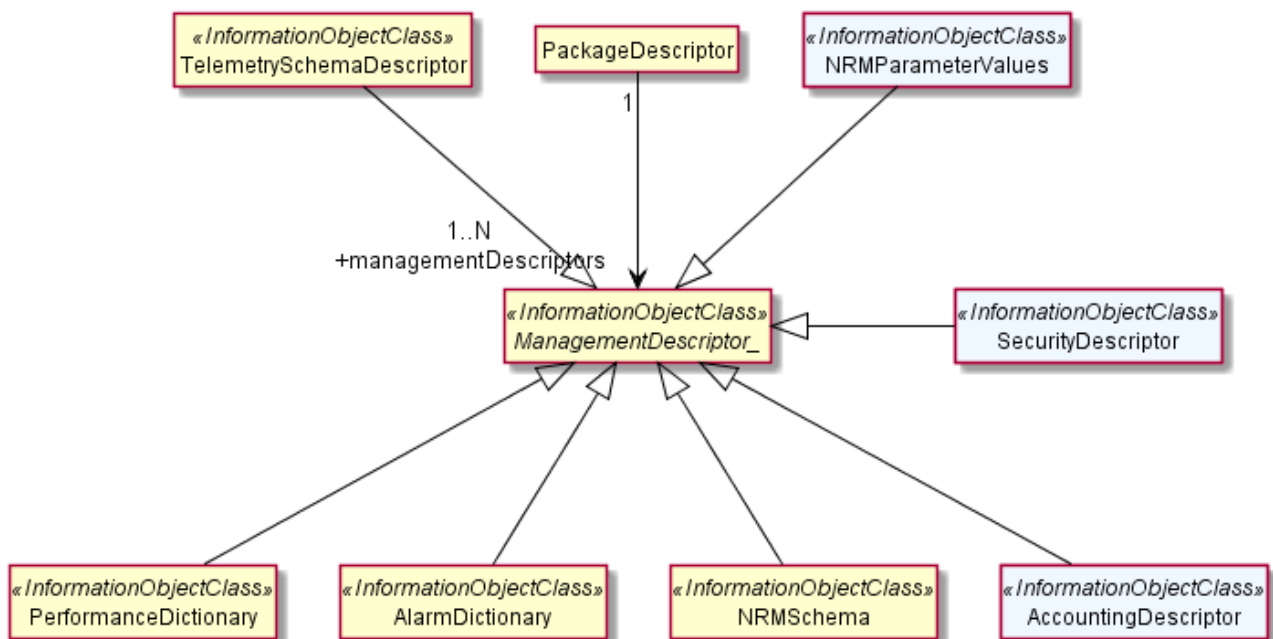


Figure 6.5.3.1-2 Management Descriptor Relationships

6.5.3.2 Alarm dictionary

Fault management consists of being able to handle alarms. An element which detects a fault may elevate the detection by notifying a management system. This is known as an alarm. Several O-RAN interfaces support the reporting of alarms. To inform the management system or SMO of those alarms and information helpful to operations personell which is not carried in the alarm event data a definition of the alarm is added to an Alarm Dictionary. The Alarm Dictionary is described in the O-RAN Information Model and Data Models Specification [2]. The Alarm Dictionary provides an interface map to map the interface event data to the definitions in the Alarm Dictionary.

6.5.3.3 Configuration descriptors

6.5.3.3.1 Telemetry schema descriptor

The NF sends many types of telemetry data to the management system and therefore there are several different structures applicable to the messages to be received. The structures of those messages are called Telemetry Schemas. The SMO needs to receive telemetry data with a “collector”. Those collectors need the schema in order to validate and accept the data from the NF. The schema definition itself is in an artifact separate from its descriptor and is normally defined by an SDO and are typically a YAML file describing a JSON schema.

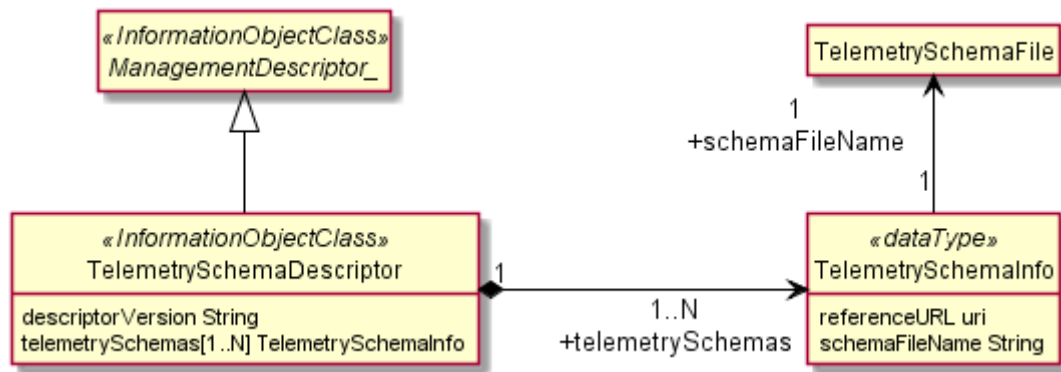


Figure 6.5.3.3.1-1 Telemetry Schema Descriptor relationships

6.5.3.3.2 NRM schema

A NF configuration model is described as a Network Resource Model (NRM), as specified in the O1 Interface Specification [3]. The schema is usually defined in YANG via several NRMSchemaFiles which reference each other. One file will be the NRMSchemaRoot. Figure 6.5.3.3.2-1 shows the relationships of the NRMSchema.

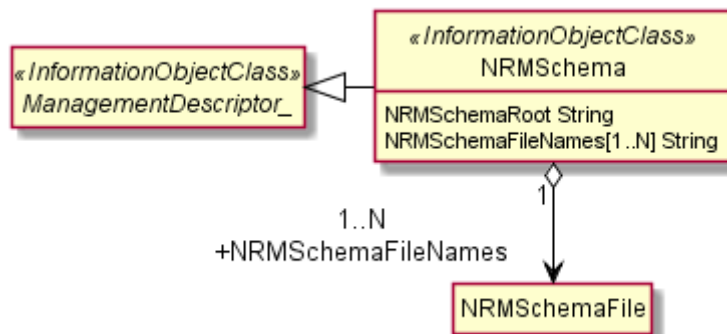


Figure 6.5.3.3.2-1 NRM Schema relationships

6.5.3.3.3 NRM parameter values

An application may be provided an partial set of parameters values which follow the NRM. The parameter values can be an XML formatted file but can also be other forms. Figure 6.5.3.3.3-1 which depicts the relationships of the parameter set to other package elements.

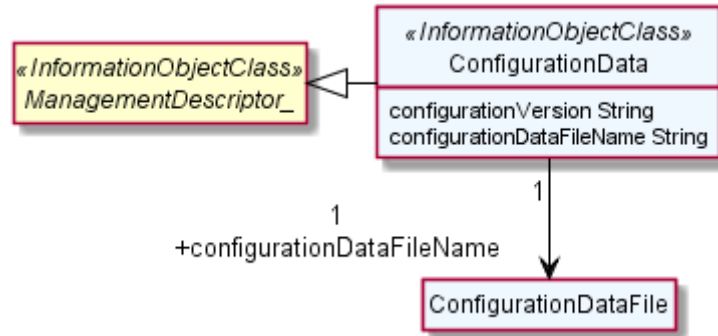


Figure 6.5.3.3.3-1 Configuration Data relationships

6.5.3.4 Accounting descriptor

In a Radio Access Network (RAN) billing records are not normally collected. However, applications may require a license to operate. A license may be tied to some level of usage. Both the license and the application come from the same vendor. Therefore, the vendor knows in advance the metadata values which are provided with a license and can also provide them in the Accounting Descriptor to automate license assignment. Usage records which define which performance metrics should be collected to validate that application usage is aligned with the license limits can also be defined in the Accounting Descriptor. Figure 6.5.3.4-1 shows the relationships of the Accounting Descriptor.

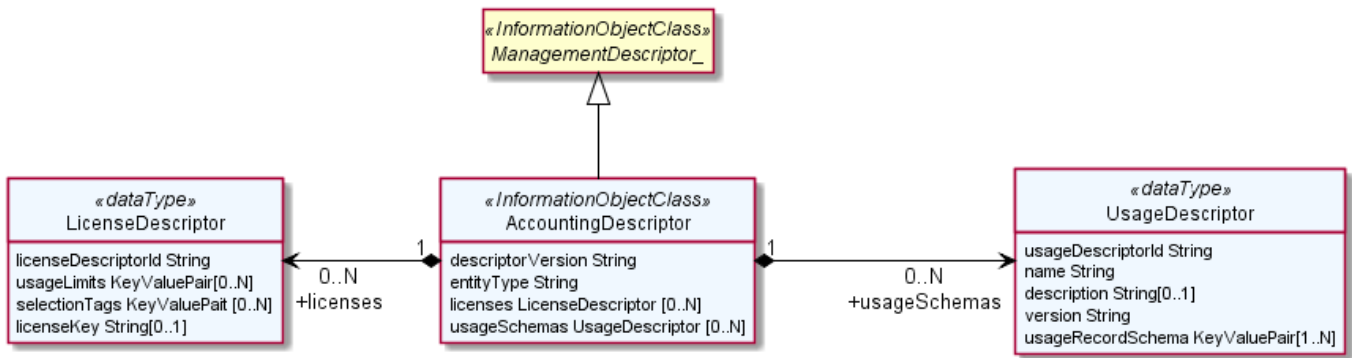


Figure 6.5.3.4-1 Accounting Descriptor Relationships

6.5.3.5 Performance dictionary

Performance management consists of being to identify what each performance metric means and how it can be used to assist operations in determining network health and/or troubleshooting when problems are detected. Several O-RAN interfaces support the reporting of performance metrics. To inform the management system or SMO of the available performance metrics alarms and information helpful to operations personell a definition of the performance metric to a Performance Dictionary. The Performance Dictionary is described in the O-RAN Information Model and Data Models Specification [2]. The Performance Dictionary provides an interface map to map the performance data to the definitions in the Alarm Dictionary.

6.5.3.6 Security descriptors

Security is done at different levels. Application security can be based on Role Based Access Controls (RBAC). In these cases, the privileges, roles, and any pre-defined users may be specified as part of the Management Security Descriptor. Other security descriptors may be added as needed to address other aspects of security management.

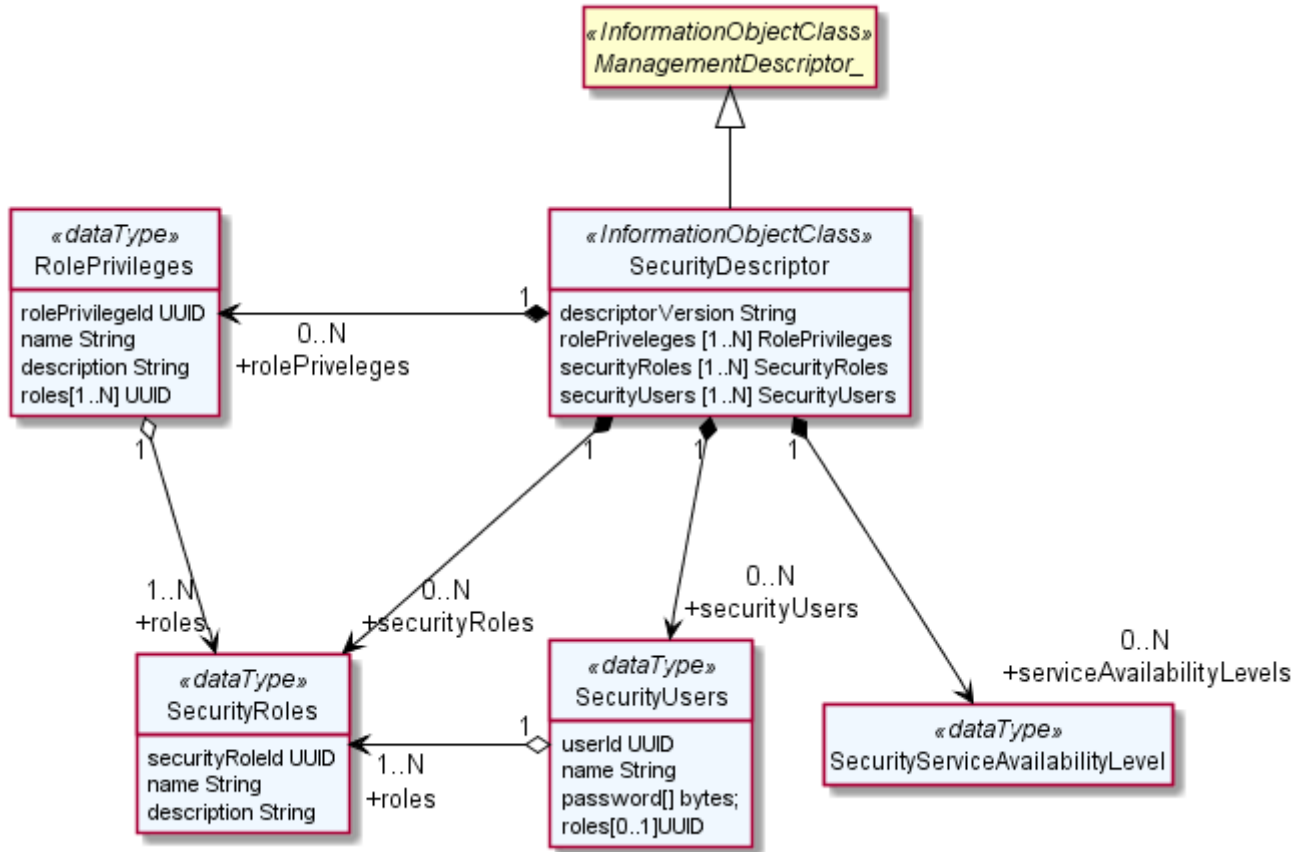


Figure 6.5.3.6-1 Management Security Descriptor Relationships

6.5.4 Requirement traceability

This aspect should be numbered 6.5.4. The section should identify applicable non-functional requirements which are defined in clause 7.2 of this specification. The traceability should be in the form of a table including the requirement identifier and an optional comment or clarification of the requirement as it applies to the use case.

Requirement	Note
REQ-OAM-CON-21	
REQ-OAM-CON-31	
REQ-OAM-CON-32	
REQ-OAM-CON-33	
REQ-OAM-CON-41	
REQ-OAM-CON-42	
REQ-OAM-CON-51	
REQ-OAM-CON-61	
REQ-OAM-CON-62	
REQ-OAM-CON-63	
REQ-OAM-CON-71	
REQ-OAM-CON-81	
REQ-OAM-CON-82	
REQ-OAM-CON-91	
REQ-OAM-CON-92	

6.6 rApp package use case

6.6.1 Description

6.6.1.1 rApp package Delivery

A solution provider has developed an rApp for deployment on the SMO platform. The rApp is delivered as a package and is capable of being lifecycle managed through the rApp management SMOS once onboarded through the onboarding SMOS.

An rApp is composed of one or more App components. Each App component contains the information needed for a set of R1 Services [4] that the rApp will produce or consume. Information relevant to a specific SMOS shall be defined within one component except for the rApp management SMOS capabilities. Information relevant to a specific rApp management SMOS capability shall be defined within one component. An App component may contain one or several files defining the content of the component. A list of the App components present in the rApp package will be contained in the App descriptor.

The following are examples of App components:

- SME – Provides a list of Service API's for rApp registration and/or discovery
- DME – Provides a list of DME Data types and data jobs for rApp registration and/or discovery
- Configuration – Provides instance configuration settings for an rApp
- Fault – Provides a list of alarms and alerts that can be raised by an rApp
- Workload – Provides the deployment descriptor, cloud native deployment artifacts and software images for an rApp

Additional App components may contain items necessary for interaction with other R1 Services. A solution provider shall include one or more App components to realize the rApp functionality.

6.6.1.2 rApp with SME component

The solution provider packages an rApp with a component to produce and consume SME related R1 Services. This includes a list of service API's it would like to produce and/or consume.

6.6.1.3 rApp with DME component

The solution provider packages an rApp with a component to produce and/or consume DME related R1 Services. This includes a list of DME types the rApp will produce and a list of DME data jobs the rApp will consume.

6.6.2 Analysis

From the use case described in the previous clause the following artifacts are identified in the rApp package:

- A manifest listing the package contents.
- An App descriptor listing the App components
- App components describing how the rApp wants to use the R1 Service, e.g.:
 - A workload component containing a Deployment descriptor (e.g. an ASD), software images and Cloud native descriptor files, e.g. Helm charts.
 - A SME component for listing R1 API's to be produced/consumed.
 - A DME component for listing Datatypes to be produced and data jobs to be consumed.

6.6.3 Information model relationships

Editor's Note: This section will be completed at a later date

6.6.4 Requirement traceability

Table 6.6.4-1 Requirements derived from this use case

Requirement	Comment
REQ-RAPP-X-01	
REQ-RAPP-X-02	
REQ-RAPP-X-03	
REQ-RAPP-X-04	
REQ-RAPP-X-05	
REQ-RAPP-X-06	
REQ-RAPP-X-07	

7 Requirements for Application Packaging

7.1 Requirement Conventions

Requirement categories are described in the “Management interface specification methodology”, clause A.1.3, In those descriptions the conceptual (CON) requirements description best matches the requirements of the Application Packaging in regard to concept, data type, relation, format, or structure.

Requirements are written based on the following template:

REQ-Label-CON-Number {Category, number} Details {Source Citation}

where "Label" is an abbreviation for the application type the requirement is scoped to, and “Number” is unique 2-digit numerical value within the label.

- COM Common to all application types

- PNF Packaging requirements specific to the Physical Network Function
- OAM Packaging requirements specific to the Management of Network Functions and/or their components
- CNF Packaging requirements specific to the Cloudified Network Function
- NFD Packaging requirements specific to the Network Function Deployment
- RAPP Packaging Requirements specific to the rApp
- XAPP Packaging Requirements specific to the xApp

7.2 Non-Functional Requirements

7.2.1 Requirements for all Application Packages

- REQ-COM-CON-01:** The Application Package shall be compliant to the packaging requirements as specified in O-RAN Security Requirements [1].
- REQ-COM-CON-02:** The Application Package shall contain information enabling an Onboarding Service to identify the type of application contained in the package (Cloudified NF package, NF Deployment package, etc.)

7.2.2 Requirements for Cloudified Network functions

- REQ-CNF-CON-01:** The Cloudified NF package shall contain a manifest listing all package contents, including URLs to external artifacts if any, and references to subordinated NF Deployment packages and a subordinated Management package, if they exist.
- REQ-NFD-CON-01:** The NF Deployment package shall contain a manifest listing all package contents, including URLs to external artifacts if any.
- REQ-CNF-CON-02:** The manifest shall contain metadata that identifies the type of artifact (e.g. software image) for each entry in the list of contents.
- REQ-CNF-CON-03:** The Cloudified NF package shall contain a Cloudified NF Descriptor.
- REQ-NFD-CON-02:** The NF Deployment package shall contain an NF Deployment Descriptor.
- REQ-NFD-CON-03:** If the NF Deployment is based on containerized workloads, the NF Deployment package shall contain one or more cloud native descriptor files or refer to them as external artifacts
- REQ-NFD-CON-04:** If the NF Deployment uses executable software, the NF Deployment package shall contain one or multiple software images, or refer to them as external artifacts.
- REQ-CNF-CON-04:** If the Cloudified NF consists of more than one NF Deployment there shall be an NF Deployment package for each NF Deployment and the Cloudified NF package shall contain references to all the NF Deployment packages.
- REQ-CNF-CON-05:** If the Cloudified NF consists of a single NF Deployment the Cloudified NF Package shall either contain a reference to an NF Deployment package or directly contain the NF Deployment package files (NF Deployment descriptor, software images, etc.) without a separate NF Deployment package.
- REQ-CNF-CON-06:** For a Cloudified NF, a management descriptor shall either be included in a Management package referred to in the manifest of the Cloudified NF package or included in one or multiple NF deployment packages.

7.2.2 Requirements for Physical Network Elements

- REQ-PNF-CON-01:** The PNF Application Package or its referenced dependent package(s) shall contain a SoftwarePackageDescriptor for every part of the PNF software architecture.

7.2.3 Requirements for Application Management Packages

- REQ-OAM-CON-21:** The Application Management Package shall contain an AlarmDictionary which identifies all alarms that can be sent from the application to the management system.
- REQ-OAM-CON-31:** The Application Management Package shall contain an NRM or its equivalent which provides the schema definition for the configuration of the application.
- REQ-OAM-CON-32:** The Application Management Package shall contain a TelemetrySchemaDescriptor which provides the schema definitions and their associated identifying URL for all messages sent from the application to the management system.
- REQ-OAM-CON-33:** The Application Management Package may contain a set of initial NRM parameter values defined for the application.
- REQ-OAM-CON-41:** When the application requires a license key then an AccountingDescriptor with at least one LicenseDescriptor shall be included in the Application Management Package.
- REQ-OAM-CON-42:** When the application requires a license key and requires specific usage data to monitor key limits then a UsageDescriptor shall be added to the AccountingDescriptor and be included in the Application Management Package.
- REQ-OAM-CON-51:** The Application Management Package shall contain a PerformanceDictionary which identifies all performance measures that can be sent from the element to the management system.
- REQ-OAM-CON-61:** When the application supports role-based access controls the default users, roles, privileges and their assignments to each other, the Application Management Package shall include a SecurityDescriptor.
- REQ-OAM-CON-62:** An Application Management descriptor may include a Security Descriptor with a service availability level, as required in O-RAN Security Requirements [1] and described in [i.4]. For CNFs and VNFs this data would be informative only.

7.2.4 Requirements for rApp Packages

- REQ-RAPP-CON-01:** The rApp package shall contain a manifest listing all package contents.
- REQ-RAPP-CON-02:** The rApp package shall contain an App descriptor identifying the App components included.
- REQ-RAPP-CON-03:** The rApp package shall contain a workload component containing a deployment descriptor, cloud native descriptor files and software images.
- REQ-RAPP-CON-04:** The rApp package may contain a SME component listing R1 API's to be produced or consumed.
- REQ-RAPP-CON-05:** The rApp package may contain a DME component listing Datatypes to be produced and data jobs to be consumed.
- REQ-RAPP-CON-06:** The rApp package may contain a Configuration component listing the configuration settings for the rApp instance.
- REQ-RAPP-CON-07:** The rApp package may contain a Fault component listing the Alarms and Alerts that the rApp instance can raise.

Annex (informative): Change history/Change request (history)

<Change history/Change request (history) is mandatory>

<Please delete the sample entries below and replace with entries for this document>

<Note that if this document will be submitted to ETSI for publication, ETSI may insert an additional “History” table on the following page. This should not be created by the authors of the current document.>

Date	Revision	Description
2024.10.09	00.00.01	Create initial O-RAN version from TS Template.
2024.12.03	00.00.02	Includes CRs: ATT-2024.09.17-WG10-CR-0003-Initial Outline and Skeleton Cleanup-v02 ATT-2024.09.17-WG10-CR-0004-Principle Data Minimization-v02
2025.11.10	00.00.03	Includes CRs: ATT-2024.10.10-WG10-CR-0005-GA Artifact Management-v04 ATT-2024.12.18-WG10-CR-0007-UC Conventions-v03 ERI-2025.03.25-WG10-CR-0164-CloudifiedNFUseCase_v05 ERI-2025.09.22-WG10-CR-0195-CloudifiedNF_InfoModel-v02 ATT-2025.01.29-WG10-CR-0008-UC PNF-v15 ATT-2025.01.29-WG10-CR-0009-UC Application Management v12 ERI-2025.10.06-WG10-CR-0203-rAppPkgUseCase-v01
2025.11.10	00.00.04	CR conflict and editorial cleanup for release
2025.11.13	00.00.05	Removed unused definitions
2025.11.25	00.00.06	Editorial comments from release review